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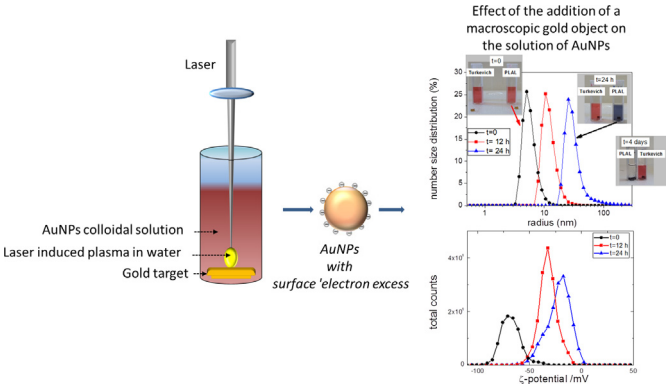
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On the stability of gold nanoparticles synthesized by laser ablation in liquids

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GRAPHICAL ABSTRACT



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ABSTRACT

"Naked" gold nanoparticles (AuNPs), synthesized in the absence of any capping agents, prepared by pulsed laser ablation in liquid (PLAL) are stabilized by negative charges. Common explanations for this phenomenon involve the presence of gold oxides and/or the anion adsorption. We have found that AuNP ablated in solutions of acids with very different oxidation power, viz. HCl, H_2SO_4, HNO_3 share the same size and ζ-potential. Although, gold oxides have p*K*as ≈ 4, the ζ-potential of AuNPs ablated in solutions with pH ⩽ 4 is always negative.

These evidences suggest that the gold oxidation and anion adsorptions have only a minor role on building the negative surface potential and we hypothesize, for the first time, that excess electrons formed within the plasma phase could charge the metallic particles. In our model, a crucial point is that the colloidal size of the NP maintains the energy of the electrons small enough to preclude chemical reactions but with a surface potential yet large enough to stabilize the AuNPs with respect to aggregation. A confirmation of the hypothesis of "electron-stabilized nanoparticles" is that either the addition of macroscopic metallic objects either the contact with a "grounded" copper wire induce the loss of charge and AuNPs aggregation.

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1. Introduction

Nanoparticles (NPs) are expected to be the key materials and building block of the 21st century. Among the inorganic NPs, those

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made of noble metals present fascinating aspects such as size-related electronic, magnetic and optical properties and interesting applications in catalysis, sensing and even medicine [1,2].

Gold nanoparticles (AuNPs), also known as gold colloids, are the most stable metal nanoparticles, and are known by centuries, for example, the book "Panacea Aurea-Auro Potabile" by F. Antonii concerning the colloidal gold dates back to 1618 [3].

Classical strategies to obtain AuNPs are based on the chemical reduction of gold salts and although the principle is very old (the formation of colloidal gold by reduction of $AuCl_4^-$ using phosphorus was described by M. Faraday in the 1857) [4] this is still an hot topic nowadays and several protocols differing in the conditions and in the nature of the reductant agent are proposed each year. The reagents, in these chemical methods of synthesis, act as reductant but also as stabilizer (capping agents) preventing (i) the undesirable endless growth of the solid particle to give a bulk gold precipitate; (ii) the aggregation of the AuNPs (a point that will be discussed in detail in the following) [5]. The capping agents bind the AuNP surface through chemical (e.g. thiols) or physical (adsorption as in the well-known case of citrate) interactions.

In some cases, the capping layer plays an important role in the practical application of the AuNPs. In particular, sensing applications often rely on the selective interaction of the analyte with the ad-hoc modified surface of the AuNP [2,5,6].

However, a purification step is always required to separate chemical products and unreacted regents from the AuNps and there are also several conditions in which the presence of molecules bound to the AuNP surface is detrimental.

In this respect, non-chemical routes to form metal NPs avoiding the presence of chemicals are very appealing.

The technique known as "Pulsed Laser ablation in liquid" (PLAL) or as "Laser Ablation Synthesis in Solution" (LASIS) is an alternative technique for forming metal nanoparticles (NPs) in solution [7]. PLAL involves focused laser pulsed irradiation of a bulk metal target in a liquid. Due to the absence of capping agents and of chemical precursors, PLAL is environmental friendly and has an easy experimental set-up.

The nanomaterials and nanostructures that can be produced are extremely varied, as well as their applications. This technique is based in four main stages [8]: Laser-matter interaction, plasma induction, cavitation bubble formation and particle release in solution. When the laser pulse of irradiance between 0.1 and 10 GW cm^{-2} irradiate the metal target in bulk liquid, breakdown process and plasma generation take place similarly to what happen in gaseous environment. The effect of background liquid on the ablation plasma consists in a stronger confinement of the plasma and in a faster transfer of energy from the plasma to the surrounding liquid [9]. The latter generates the cavitation bubble which, in turn, is a dynamic system characterized by an expansion phase and a collapse phase after the bubble reaches the maximum radius. During the collapse of the bubble, NPs are released in the liquid producing a stable colloidal solution.

The interest in a capping agents-free synthesis of metal NPs has stimulated fundamental studies on the impact of the laser-induced plasma induction and of the cavitation bubble generation in the nucleation processes leading to the formation of NPs. At variance, the reasons of the stability of the NPs generated by PLAL is still an open issue. It is worth emphasizing that the stabilization of noble metal colloids is challenging. The inter-particles attraction due to the van der Waals forces is proportional to the so called Hamaker constant (H) that is very large in the case of noble metals. For example, considering the water as dispersing medium, the Hamaker constant for gold is ~100 k_bT while for polystyrene it is ~2 k_bT [10].

The attractive potential V_A between two identical spheres due to van der Waals interactions can be approximated as [20–22]:

$$V_A = -\frac{H}{6}\left[\frac{2R^2}{h^2-4R^2}+\frac{2R^2}{h^2}-ln\left(\frac{h^2}{h^2-4R^2}\right)\right] \quad (1)$$

where R is sphere radius and h is the edge-to-edge distance between the spheres.

According to the above equation, the attraction between AuNPs is 50-fold larger than between latex particles of the same size and, in the absence of a strong repulsive inter-particle force, the fate of a AuNPs dispersion should be an immediate, irreversible coagulation.

Chemically synthesized NPs are usually stabilized by a surface layer of capping agents. These molecules, beside avoid the infinite growth of the solid phase, impart a repulsive inter-particle interactions. Basically, the capping layer induces a repulsion between NPs due to steric or electrostatic interactions (or due to a combination of both) [5,20].

Strikingly, PLAL performed in pure solvents, i.e. in the absence of any capping agents, leads to the formation of very stable "naked" NPs that are long standing for months [12]. Several papers have assessed that the metal NPs prepared by PLAL have a large negative zeta-potential (ζ) and therefore their stability should be electrostatic in nature and it is due to the presence negative surface charges.

The question is what is the origin of these surface charges? The case of the gold NPs is emblematic. Gold is a noble metal that is inert under most conditions but, as a matter of fact, AuNPs prepared by PLAL in pure water have a large negative zeta-potential.

Some authors ascribe the negative charge to the presence of partially oxidized gold species (AuO^- and $AuCO_3^-$) [13,14]. However, the presence of oxidized gold species on the surface of NPs prepared in pure water has been questioned by others [15]. A recent investigation, suggests that the negative surface charge could be dominated by ion adsorption and not by surface oxidation [16]. In general, a fair comparison among these different results is complicated because the experiments have been performed using different setups and it is known that the ablation laser pulse duration, fluence and wavelength all impacts the mechanism of NPs production.

To clarify these issues, we have undertaken a systematic study synthesizing AuNPs through PLAL using as liquid phase deionized water and aqueous solutions of salts, acids and bases.

2. Materials and methods

The reagents used for the solutions preparation are reported below: the potassium chloride (RPE grade, CAS number 7447-40-7), the hydrochloric acid (37% RPE grade, CAS number 7647-01-0), the sodium bicarbonate (RPE grade, CAS number 144-55-8) were purchased from Carlo Erba Reagents; the sodium hydroxide (ACS reagent, CAS Number 1310-73-2), the sodium citrate tribasic dihydrate (ACS reagent > 99%, CAS Number 6132-04-3), the sodium carbonate anhydrous (ACS reagent > 99.95%, CAS Number 497-19-8), the sodium phosphate monobasic monohydrate (ACS reagent > 99%, CAS Number 10049-21-5), the sodium phosphate dibasic anhydrous (ACS reagent > 98%, CAS number 7558-79-4), the sodium acetate anhydrous (ACS reagent > 99.0%, CAS number 127-09-3), the bromophenol blue (ACS reagent, CAS number 115-39-9) were purchased from Sigma Aldrich.

2.1. Synthesis of AuNPs via PLAL

The experimental setup for PLAL consists of nanoseconds Nd-YAG laser (Quanta System Thunder) operating with fundamental (1064 nm), having a laser frequency of 10 Hz and a nominal pulse duration of 8 ns as described in a previous work [17].

Fig. 1 shows a sketch of the experimental set-up. For the production of AuNPs in water, the laser was focused on a gold target immersed in a vessel filled with 25 ml of water or salt solution by using of a 5 cm focal lens; the irradiance was of 71 GW/cm^2, and the laser-ablation time of 6 min.

These experimental conditions were chosen on the basis of reproducibility, size distribution, and stability in time. The gold target was purchased from Kurt J. Lesker Company (99.999% pure pellet). All the solutions were prepared in ultrapure Milli-Q water. The AuNPs at pH = 3–10 were prepared by ablating the gold target in suitable buffer (50 μM of buffer) or acid aqueous solutions. For pH 6–7–8 KCl and monobasic/dibasic phosphate buffers were employed while for the solution at pH 10 NaOH and bicarbonate/-carbonate buffer were used. For the experiment at pH 4 and 3 the pH was fixed by the presence of strong acids HCl, HNO_3 or H_2SO_4 at suitable concentration.

The here reported results for AuNPs synthesized in different solutions have been replicated at least three times (i.e. different synthesis). When the ablation parameters are kept fixed, the nature of the liquid phase has a small influence on the size and concentration of the AuNP and the whole set of AuNPs examined in this work presents an average diameter of 13 ± 3 nm, and concentration of 0.08 ± 0.02 mg/mL.

2.2. Synthesis of AuNPs stabilized by citrate

Chemical synthesis of gold nanoparticles stabilized by citrate anions has been performed following the classical Turkevich method [18]. Briefly, 2 mL of 1 wt% trisodium citrate aqueous solution were rapidly injected into 20 mL of $HAuCl_4$ 1 mM boiling solution under vigorous stirring. The formation of AuNPs is indicated by the color of the solution, which turns from the original pale yellow to burgundy. After 15 min the solution was cooled down by means of a water-ice bath. All the chemicals were from Sigma-Aldrich.

For comparison AuNPs have been also synthesized by means of PLAL in a solution of citrate (2 mM) with the PLAL conditions described above.

2.3. AuNPs characterization

A first fast characterization of the produced NPs has been carried out acquiring the Surface Plasmon Resonance (SPR) spectra of the solution using an Ocean Optics (USB2000 + XR) spectrometer with a light source (Mini Deuterium Halogen Light Source DT-Mini-2-GS). The calibration lines described in [17], were used to estimate the concentration and the diameter of the produced AuNPs.

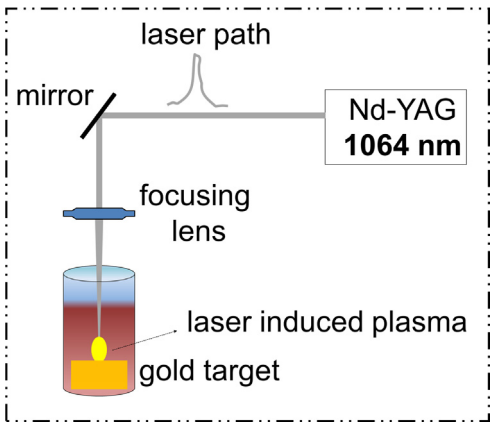


Fig. 1. Sketch of the experimental set-up.

A further characterization of the produced AuNPs was carried out using the Dynamic Light Scattering and ζ-potential measurements using a Zetasizer-Nano ZS from Malvern Instruments operating with a 4 mW He-Ne laser (633 nm wavelength) and with the cell holder maintained at 25 °C by means of a Peltier element.

The instrument measures hydrodynamic size by dynamic light scattering DLS and the ζ-potential by laser Doppler electrophoretic through phase analysis light scattering, LDE-PALS [19].

DLS measurements were performed in backscattering at fixed detector angle of 173° (NIBS™) while the ζ-potential measurements were performed using forward scattering (17°) setup in capillary cells.

LDE-PALS furnishes the electrophoretic mobility of the AuNPs and, due to the very low ionic strength of all the samples, the ζ-potential has been evaluated from the mobility using the Hückel equation (see the following Section 3):

DLS data were collected leaving the instrument free to optimize the instrumental parameters (attenuator, optics position and number of runs). Usually the time autocorrelation function (ACF) of scattered light intensity was the average of 10–12 consecutive runs of 10 s each.

The size distribution by number was recovered, using the software implemented by the manufacturer, by taking the inverse Laplace transform of the ACF and subsequent application of Stokes-Einstein equation assuming the viscosity of the water solution at 25 °C and the complex refractive index of bulk gold.

2.4. pH measurements

Routine pH measurements have been performed using the pH-meter Crison Instruments Basic 20 equipped with the electrode 5011T.

Spectroscopic determination of pH in acidic solutions has been carried out probing the peak of the basic form of bromophenol blue at 591 nm (a wavelength where the SPR of the AuNPs produced by PLAL is almost negligible). The amount of bromophenol blue added is chosen to have an absorbance of 0.5 in solution at pH = 4.1 (1 cm path length). To determine the change in pH upon ablation, the bromophenol blue was added to the solution after the ablation and to an equal volume of the same solution that has not used for the ablation and the respective pH-values have been estimated using a calibration curve prepared using acetic acid/acetate buffer of 0.01 M solutions covering a pH between 3 and 6. The absorbance spectra for the spectroscopic pH determination have been acquired by using as a reference spectrum the AuNPs solution, in order to put in evidence the bromophenol blue peak variations.

3. Colloidal stability: Basic background [5,10,11,20]

The ubiquitous van der Waals interactions set an attractive potential between particles of the same material. Only in the presence of a repulsive interaction that counterbalance these attractive interactions a particle suspension remains stable. Inter-particle repulsion takes place in solution if the particles are charged.

Macroscopically, any charge on the surface of a colloidal particle must be balanced by the opposite charge of the counterions present in the solution. Such a situation is usually described in terms of an electrical double layer: a layer of a given charge is on the surface of the particle and the opposite charges are localized on a more or less diffuse “cloud” in the solution. The surface potential and the ionic strength of the solution dictate how diffuse is the counterions layer.

According to the DLVO theory, developed independently by Derjaguin and Landau and by Verwey and Overbeek, a colloidal dispersion is stable if there is a balance between the repulsion from

the double-layer and the attraction due to van der Waals forces. The total interaction energy function, V_T, is the sum of the attractive potential energy, V_A, and the repulsive potential energy, V_R.

$$V_T = V_A + V_R \tag{2}$$

For spherical colloids the attractive potential V_A can be approximated by the Eq. (1) above.

The ionic atmosphere around the charged spherical particles forms the Gouy-Chapman diffuse double layer that extend on distance of the order of the so called Debye's length λ_D defined as

$$\lambda_D = \sqrt{\frac{\varepsilon\varepsilon_0 k_b T}{e^2 \sum_j z_j^2 c_j}} \tag{3}$$

where ε is the relative dielectric constant of the medium, ε_0 the vacuum permittivity, e the elementary charge, k_bT the thermal energy and the summation is twice the solution ionic strength, with z_j and c_j being the valence and the concentration of the j-th ion respectively.

The repulsive potential between charged spheres cannot be solved analytically but different approximations have been found by Verwey and Overbeek depending on the ratio between the particle radius and the Debye's length [21]. The case relevant to particle prepared by PLAL is that of small particles in solution at low ionic strength ($R/\lambda_D \ll 1$). To qualitatively figure out the factors affecting the stability of AuNPs under the condition $R/\lambda_D \ll 1$, we will use the simple approximation proposed "where no great precision is required" by Verwey and Overbeek [20,21]:

$$V_R \approx \frac{4\pi\varepsilon\varepsilon_0 R\psi_0^2}{\frac{h}{R}+2} exp\left(-\frac{h}{\lambda_D}\right) \tag{4}$$

where ψ_0 is the potential at the particle surface.

The total interaction energy can be written as

$$V_T = V_A + V_R = -\frac{H}{6}\left[\frac{2R^2}{h^2-4R^2} + \frac{2R^2}{h^2} - ln\left(\frac{h^2}{h^2-4R^2}\right)\right] + \frac{\varepsilon\varepsilon_0 R\psi_0^2}{\frac{h}{R}+2} exp\left(-\frac{h}{\lambda_D}\right) \tag{5}$$

V_A and V_R have different dependences on h, where V_A becomes more negative as the inter-particle distance h decreases, while V_R is positive but decays exponentially with h. The attractive term V_A dominates at very long or very short distances, while the repulsive term V_R can dominate, providing that the pre-exponential term is large enough, only for distances that are of the same order of the Debye's length ($h \approx \lambda_D$). Thus, the contributions of V_A and V_R to V_T vary with h. Consider the case of low and high screening. For low electrolyte concentrations or low screening, the Debye length λ_D is long, resulting in a high V_R. This results in an energy barrier in V_T which must be crossed if the particles were to come into contact with one another. As an exemplification, in Fig. 2 are plotted the predictions of Eq. (5) for gold sphere with a radius of 5 nm and a surface potential $\psi_0 = -60$ mV immersed in solutions at different ionic strengths. The lowest ionic strength attainable in water (due to the water self-dissociation) is 10^{-7} M (black curve). A repulsive potential barrier that reaches more than 7 k_bT at the maximum is present and hinders the particles approach each other and reach the region where V_A dominate.

For higher electrolyte concentration, the Debye screening length reduces, and this reduces the high of the potential barrier; this is also shown in Fig. 2 where the V_T in the case of ionic strength of 10^{-5} M (relevant for deionized water in equilibrium with atmospheric CO_2). The higher the ionic strength is the lower the maximum of V_T is until it equals the thermal energy (this takes place for ionic strength around 10^{-3} M in Fig. 2). Under this condition the particles are not prevented from coming in contact with one another and they reach the regime where V_A dominates and irreversible coagulation takes place. The minimum concentration of salt enough to induce irreversible coagulation is known as the critical coagulation concentration, CCC, and is function of the ions valence and also of the nature of the particles. For example in Fig. 2 it is also shown the interaction potential between latex particles of the same size at ionic strength = 10^{-3} M. It can be seen how at this ionic strength, that is the CCC for the gold particles, the latex spheres are yet stable with a barrier of three k_bT.

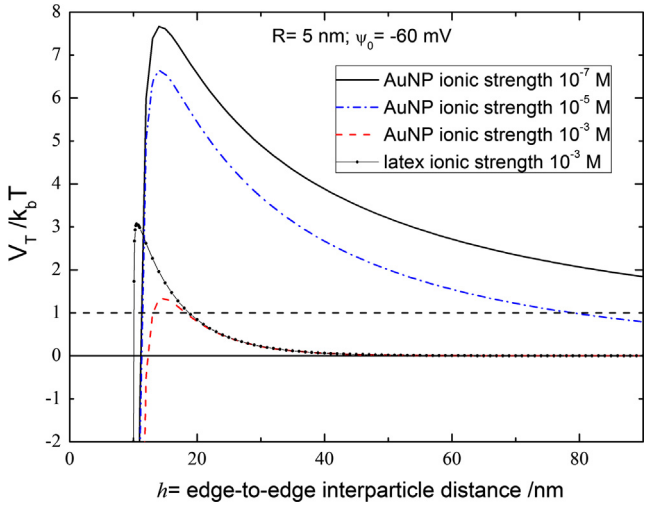


Fig. 2. Inter-particles potential as function of the distance according to Eq. (5) for particles at constant surface potential (−60 mV) and particle radius (5 nm). Lines are the calculation for gold particles in solutions at different ionic strength. The line with dots refer to latex (organic polymer) particles with the same radius and surface potential in solution at ionic strength = 10^{-3} M. the dashed horizontal line denotes the thermal energy level (1 k_bT).

The electrostatic potential generated by the charges on a colloidal particle can be probed by electrophoresis experiments [11,19,20]. The laser Doppler electrophoresis (LDE) technique used in this work, strictly probes the electrophoretic mobility (u_e) of the particle. In turn u_e can be related to the ζ-potential that is the potential at the plane where the liquid velocity relative to the particle is zero (the slipping plane). Note that only if the slipping plane is located at the particle surface, ζ becomes equal to electrical potential at the surface of the NP but, being the only experimental observable that reflect the surface charges, the ζ-potential is usually taken as a good approximation of the surface potential. Considerable care is required in the correlation between mobility and ζ-potential because the Henry's equation, $u_e = \frac{2}{3}\epsilon\epsilon_0\frac{\zeta}{\eta}f\left(\frac{R}{\lambda_D}\right)$, rests on (beside the viscosity η) on the Henry function $1 \leqslant f\left(\frac{R}{\lambda_D}\right) \leqslant \frac{3}{2}$ that depends on the ration between particle size and Debye's length. The Smoluchowski's limit ($f = 1.5$) holds for $R/\lambda_D > 1$ meaning relatively large particles in high salt solutions. In the opposite case ($R/\lambda_D \ll 1$), relevant to particle prepared by PLAL, the Hückel limit ($f = 1$) is more appropriate and, accordingly, it has been used to evaluate all the ζ-potential values in the present work.

4. Results and discussion

The stability of AuNPs prepared by PLAL is due to the electrostatic double layer repulsion. Indeed we found that these particles aggregates when the ionic strength is high enough, with a CCC ≈ 0.05 M for monovalent salts. It is well assessed that AuNPs

prepared by PLAL have a large negative ζ-potential meaning that the NP must be negatively charged [12,13,15]. Being "naked" particles without capping agents or stabilizers the origin of these surface charges is puzzling.

In practice, colloidal particles in solution are always charged. There are three basic mechanisms that generate charged interfaces: (i) the colloids are made by intrinsically charged species (as clay where aluminum replaces silicon giving rise to an excess of negative charges); (ii) an intrinsically uncharged colloid acquires charges from a species adsorbing from solution; (iii) titratable groups are present on the surface making their charge pH dependent [11].

The nature of gold NPs excludes the mechanism (i) above and, as a matter of fact, negatively charged AuNPs are formed by laser ablation also in pure water (where the presence of charged contaminants should be negligible). It is thus natural that the most promising candidate for the stability of the AuNPs has been the presence of amphifunctional groups that can be deprotonated forming negative charges. In a seminal work by Sylvestre et al. [13], traces of oxidized gold on the surface of AuNPs synthesized through PLAL in deionized water have been inferred from the analysis of XPS spectra and the TOF-SIMS spectra showed traces of $Au\text{-}O^-$ and $Au\text{-}CO_3^-$ moieties. Accordingly, the authors have attributed to the carbonation and to the hydroxylation and subsequent deprotonation of these Au-O groups the negative charging of the nanoparticles.

However, this is not necessarily the cause of the negative charges on AuNPs under all the ablation conditions. In a subsequent work by Buntine and coworkers [15], the AuNPs synthesized in deionized water using, a different PLAL setup, do not show traces of oxidized gold in the XPS spectrum but are yet very stable.

The difference in the surface chemical composition is likely due to the differences in ablation laser pulse, fluence and wavelength but remains the fact that the stability of the NPs is unaffected by the presence gold oxides on the surface of NPs.

There is another point that suggests a minor role for gold oxides. The hydroxyl groups linked to gold have been studied on the surface of bulk gold and are known to be amphifunctional in the sense that they can either release or bind protons. The relevant reactions of the —AuOH groups are

$$—AuOH_2^+ \overset{K_{a1}}{\leftrightarrow} —AuOH + H^+ \qquad \text{(reaction1)}$$

$$—AuOH \overset{K_{a2}}{\leftrightarrow} —AuO^- + H^+ \qquad \text{(reaction2)}$$

Meaning that at $pH > pK_{a2}$ the AuNPs stabilized by a layer of gold hydroxide will carry a negative charge that will decrease upon reduction of pH until at $pK_{a1} < pH < pK_{a2}$ the NPs are neutralized. Note that for further decrease in pH the particles should achieve a *positive charge* for $pH < pK_{a1}$. This is exactly what has been observed on macroscopic gold plates by means of streaming-potential measurements by Duval et al. [23]. They have found that the ζ-potential of the gold plate is negative for pH > 4.5, becomes null at $pH \approx 4$ and is positive for pH < 4. The study allowed the determination of the acid constants of the reactions 1 and 2 above and specifically $pK_{a1} = 3.9$ and $pK_{a2} = 4.2$.

The behavior reported in the case of AuNPs having oxidized gold on the surface is, however, very different. The ζ-potentials reported by Sylvestre et al. [13], in the case of gold nanoparticles produced in 10 mM aqueous NaCl solutions at different pH values ranging from 3 to 10, are always negative and only a decrease in the absolute value of ζ is found below pH = 5.8.

To clarify such an issue we have undertaken a systematic investigation on the effect of the solution pH on size and ζ-potential of the AuNPs prepared by means of PLAL.

To this purpose, gold nanoparticles have been prepared in 10^{-4} M solutions of HCl, KCl and NaOH. Representative spectra of these particles, synthesized by PLAL in solutions having the same ionic strength and ranging 6 pH-units, are shown in Fig. 3A. In all the cases despite the large difference in pH the SPR peak are close to 520 nm indicative of small NPs with a radius of the metallic core (R_c) around 5 nm. The DLS confirms this evidences as shown in the number size distributions reported in Fig. 3B that are focused in the region of small hydrodynamic radii (R_h) < 10 nm. In this respect, we should remark that the hydrodynamic radius obtained from DLS data takes into account all the contributions to the friction of the particles undergoing diffusion, comprising the adsorbed solvent layers and the coating (if any) [24].

The laser Doppler electrophoresis experiment performed by means of phase analysis light scattering (PALS) allows, when the ionic strength is low enough, to reconstruct the distribution of

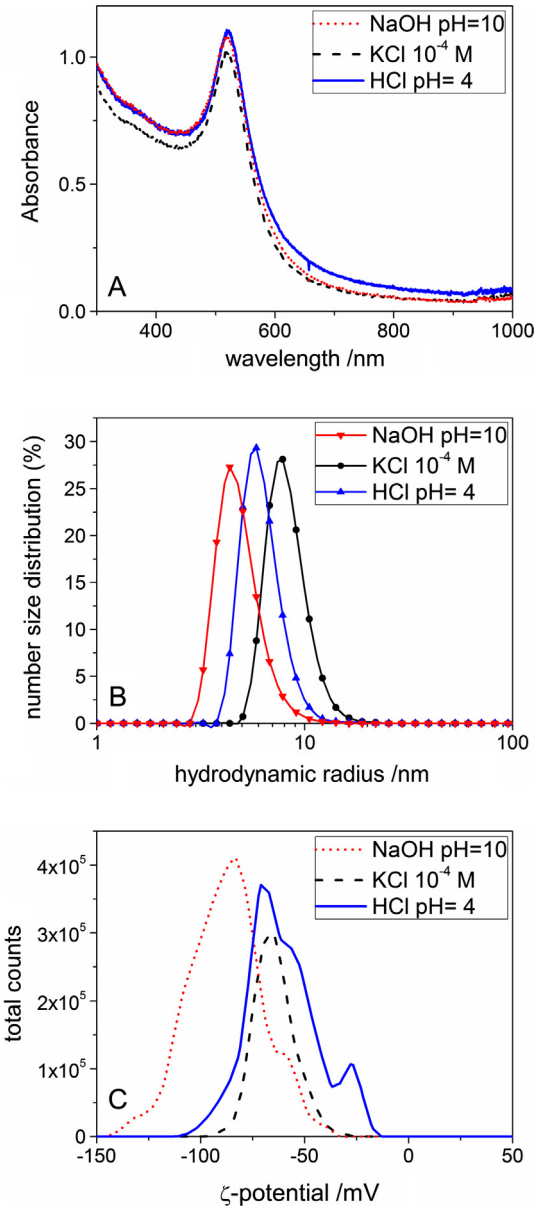


Fig. 3. Absorption spectrum (panel A), number size distribution from DLS (panel B) and ζ-potential distribution (panel C) for AuNP ablated in solutions with the same ionic strength (10^{-4} M) and different pH namely: pH = 10 (red lines); KCl (black lines); pH = 4 (blue lines). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

ζ-potential instead of the single average value. Representative ζ-distribution are shown in Fig. 3C for AuNP ablate in different liquid solutions. All the particles are negatively charged and, although the ζ-potential corresponding to the maximum in the ζ-distribution becomes more negative passing from −60 mV in HCl at pH = 4 to −88 mV in NaOH at pH ≈ 10, the differences are small and the ζ-distribution largely overlaps each other.

Stable AuNPs have been successfully synthesized by means of PLAL also in solutions of acids (HCl, H_2SO_4, HNO_3) at pH = 3 and in this case, as well, we observe small sizes ($R_c \approx 5$ nm) and ζ-potential ≈ −60 mV (see Table 1).

A fair discussion of the effect of the pH on the ablation solution is complicated by the presence of dissolved CO_2 (the ablation is performed in open vessels) and because the pH of the solution changes upon ablation. This last effect is small (tenths of pH) but is measurable in acidic solutions (above pH 4 the fluctuation in the CO_2 obscure any effect). To increase the accuracy, the change in the pH upon PLAL has been determined spectrophotometrically by probing the band at 591 nm of the basic form of the pH-indicator bromophenol blue.

The spectra shown in Fig. 4 refer to bromophenol blue dissolved in water acidified with HCl and in the same solution after PLAL. It is clear that the PLAL process induces a sizeable increase of the band at 591 nm and a decrease of the band at 435 nm, both indicative of an increase in pH. Comparison with the calibration curve allows the evaluation of the pH-values: before ablation pH = 3.44 ± 0.01 and immediately after the ablation pH = 3.62 ± 0.01. Reassuringly the pH of the solution doesn't change after one day of aging.

The above described results, can be hardly attributed to a significant contribution of gold hydroxyls to the NP charge. In that case the ζ-value of the AuNPs should reverse its sign crossing pH = 4 and this is not observed. Furthermore, the *increase* in pH upon ablation at pH = 5 is difficult to attribute to the formation of an acidic species. Also the invariance of the ζ-value among AuNPs ablated at pH = 3 in solutions of HCl, H_2SO_4 and HNO_3, suggests a marginal role of the surface oxides considering that HNO_3 is an oxidant much stronger than the other two acids.

On the other hand, we observe essentially the same ζ-value at constant pH independently from the ions present in solution (experiments with different buffers and with acids/bases plus salt give the same results), and this rules out a fundamental role of ion adsorption on building the charge on the NPs. Further interesting point is the formation of stable, negatively charged NPs also at pH 3 and 4 because this excludes any role of carbonates in the stabilization of AuNPs.

Having ruled-out surface deprotonation and anion binding as the origin of negative charges on the AuNPs the next candidate is the most abundant negatively charged species found in the laser-induced plasma: the electron. Recent studies (see for example [25]) are suggesting that the NPs are formed in the liquid confined plasma as a results of the excess of electrons in the plasma phase, that charges negatively the solid clusters and induces their growth by ions attraction. When the number of ions becomes negligible because of plasma recombination and particle formation, the size of the NP depends on the plasma temperature as a consequence of the evaporation/condensation equilibrium, but, as it commonly occurs in plasma systems, the produced NP could save its negative charge.

Of course electrons that are generated during the laser ablation process are high-energy, highly reactive and short-living species. However, the presence of a metallic domain in an high dielectric constant medium could drastically change the state of the affair. The free-energy of Z extra electrons on a metallic sphere of radius R equals the Born energy to charge the sphere itself that is [10]:

$$\Delta G = \frac{Z^2 e^2}{8\pi\varepsilon\varepsilon^\circ}\frac{1}{R} = Z^2\frac{l_B}{2R}KT \qquad (6)$$

In the second equality the Born energy is expressed in terms of the ratio between the Bjerrum length l_B (i.e. the separation at which the electrostatic interaction between two elementary charges equals the thermal energy KT) that in water is 0.7 nm and the particle diameter. According to the above equation, the free energy of one excess electron staying on a NP of 5 nm in radius immersed in water is only 0.07 KT a cost that can be easily compensated by the thermal bath. However, such a free energy cost scales as the square of the number of charges and thus above a certain critical amount of electrons (Z) the reaction of the electrons with other species become competitive.

In pure water, the only species that can be reduced by the excess electrons are the hydrated H^+ ions (strictly the hydronium should be denoted as H_3O^+ but for the sake of simplicity we will use the notation H^+) that can react with a metallic nanoparticle bearing Z electrons to give gaseous hydrogen according to the following reaction:

$$NP^{Z-} + H^+(aq) \rightarrow NP^{(Z-1)} - + 1/2H_2(g) \qquad (\text{reaction3})$$

Were NP^{Z-} and $NP^{(Z-1)-}$ denotes the NP bearing Z and Z−1 electrons, respectively.

Considering that the gold is solid state, the reaction free energy ($\Delta_{rea}G$) per H^+ molecule is

$$\Delta_{rea}G = \Delta_{rea}G^\circ + \frac{1}{2}KTlnP_{H_2} - KTln[H^+]_{sup} \qquad (7)$$

Since the ion distribution around a charged particle is not uniform, the relevant H^+ concentration is that at the surface of the metallic particle $[H^+]_{sup}$.

Considering that the H_2 concentration in the atmosphere is about 530 ppb [26], the $P_{H_2} \approx 0.0077$ atm and the reaction free energy can be written as:

$$\Delta_{rea}G = \Delta_{rea}G^\circ - 2.435KT + 2.3KTpH_{sup} \qquad (8)$$

Table 1
Metallic core (R_c) and hydrodynamic (R_h) radius and ζ-potential of AuNPs ablated in acidic and in KCl solutions.

Ablation medium	R_c/nm (from SPR)	Rh/nm (from DLS)	ζ-potential/mV
HCl pH = 3	6 ± 3	7 ± 1	−57 ± 3
HNO_3 pH = 3	6 ± 3	6 ± 3	−60 ± 3
H_2SO_4 pH = 3	6 ± 3	7 ± 2	−66 ± 6
KCl 10^{-4} M	5 ± 3	6 ± 2	−67 ± 3

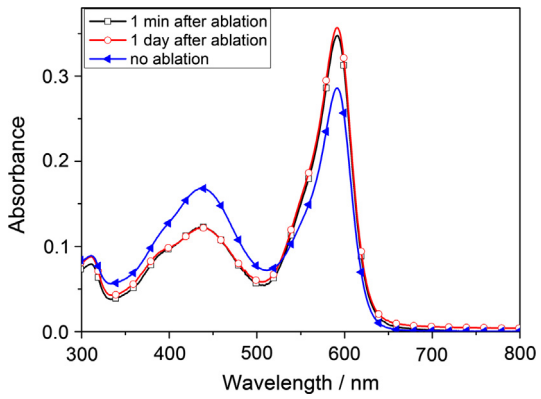


Fig. 4. Absorption spectrum after addition of the pH-indicator blue bromophenol to acidic water (blue) and to the same acidic water after PLAL (black). Also it is shown the spectrum of the latter solution after one day (red). The spectra in the presence of AuNPs (i.e. after PLAL) have been acquired by using as a baseline the AuNPs solution. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

where pH_{sup} denotes the pH value at the nanoparticle surface. The $\Delta_{rea}G°$ is the reaction free energy for products and reagents in their standard state and can be evaluated from their formation free energy. The free energy of formation of $H^+(aq)$ and H_2 in their standard state (1 M and 1 atm of pressure) is null by definition and so is their contribution to the $\Delta_{rea}G°$. The difference in free energy of formation of the nanoparticles differing only in the number of electrons is given by the change in the electrostatic energy of a metallic sphere of fixed radius R and different number of electrons and according to Eqs. (5) and (8) can be written as.

$$\Delta_{rea}G = \frac{l_B}{R}\frac{(1-2Z)KT}{2\left(1+\frac{R}{\lambda_D}\right)} - 2.435KT + 2.3KTpH_{sup} \quad (9)$$

The surface concentration of H^+ depends on the surface potential according to the Boltzmann's distribution [10,11]:

$$[H^+]_{sup} = [H^+]exp\left(-\frac{e\psi_0}{KT}\right) \quad (10)$$

In the above equation, $[H^+]$ is the concentration in the bulk where the potential vanishes ($\psi = 0$). The surface potential ψ_0 and the charges on the nanoparticles (i.e. Ze) are related but a closed form can be found only in the case of low surface potential. Within this approximation and in the limit $R/\lambda_D \ll 1$ the surface potential is given by the simple relation:

$$\psi_0 = Z\frac{l_B}{R}\frac{KT}{e} \quad (11)$$

Combining the Eqs. (9)-(11) and considering the equilibrium ($\Delta_{rea}G = 0$) one obtains:

$$0 = \frac{l_B}{R}\frac{(1-2Z)}{2\left(1+\frac{R}{\lambda_D}\right)} - 4.87 + 2.3pH + Z^2\frac{l_B}{R} \quad (12)$$

The positive root of such a quadratic equation gives the dependence of the excess electrons (Z) on the pH. The surface potential ψ_0 can be evaluated from Z according to Eq. (11) and the result of these calculation can be compared with the outcome of electrokinetic phenomena such as the ζ-potential. The calculated surface potential ψ_0 is compared with the measured the ζ-potential in Fig. 5A and the calculated excess charges are plotted as a function of the pH of the ablation solution in Fig. 5B.

Despite the approximations implicit in Eq. (11) (that holds for ψ_0 small enough to linearize the spherical Poisson-Boltzmann relation between the potential and the ion distribution i.e. $\psi_0 \ll 25$ mV), there is a remarkable agreement between the calculated ψ_0 and the measured ζ-potential. This strongly suggests that, at least under the conditions of PLAL explored in this work, the negative charge on the AuNPs could be due to excess electrons that have been produced during laser ablation induced plasma. A crucial point is that the colloidal size of the NP maintains the energy of the electrons small enough to preclude chemical reactions but with a surface potential yet large enough to stabilize the AuNPs with respect to aggregation. If this is true, the addition of a macroscopic metallic object to the system represents a stringent test for the hypothesis of electron-induced colloidal stabilization because the bulk metal should act as sink for the electrons on the AuNPs (according to Eq. (5), when the object size tends to infinite, the energy cost of charging becomes negligible). Following such a line of reasoning, we have added a small piece of metallic gold to the solution of AuNPs (prepared by PLAL) probing the SPR band, the hydrodynamic size and the ζ-potential as a function of time. In parallel the same experiment has been conducted on AuNPs synthesized *via* reduction of the $HAuCl_4$ by citrate according to the Turkevich protocol. The experiments have been conducted on five

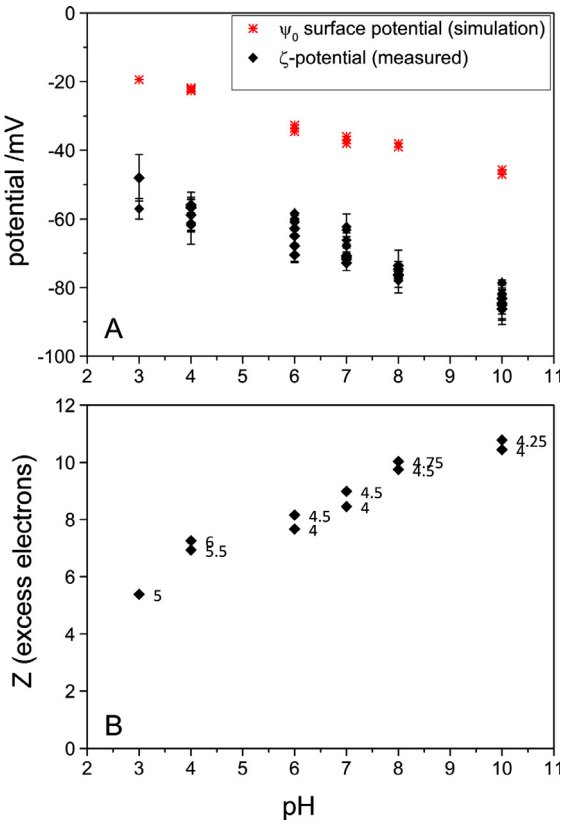


Fig. 5. (Panel A) solid diamonds are the experimental ζ-potential of AuNPs ablated in buffers at different pH-values. Different points refer to different ablation experiments. For each ablation five measurement of the ζ-potential have been performed and the point and the error bars are the mean and the standard deviation of these replicates. Red asterisks are the surface potential calculated for AuNPs according to Eqs. (11) and (12). The radii used in the calculation are the same determined experimentally from the SPR for the AuNPs ablated (for this reason there are more than one value for each pH). (Panel B) The number of excess electron calculated for AuNPs ablated at different pHs according to Eq. (12). For the sake of clarity only the outcomes for two radii is shown. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

different PLAL preparations of AuNPs and representative results are shown in Fig. 6.

In the central panel of Fig. 6 are also shown the pictures of the solutions before addition of bulk gold, after 24 h from the gold addition and after 4 days. It is evident that, only in the case of AuNPs prepared by PLAL, the addition of a metallic object turns the solution to a blue color in one day and induces the irreversible aggregation of the nanoparticles on longer timescale (4 days). The change of color, reflects the shift of the SPR band towards longer wavelengths (Fig. 6A) as a consequence of the stepwise aggregation of the NPs. This is confirmed by the increase of the hydrodynamic radius probed by DLS (Fig. 6B). Parallel to this process the ζ-potential becomes less and less negative (Fig. 6C). In this experiment there are not changes in the chemical composition of the solution (ionic strength and pH are not perturbed) so we ascribe the drop in the absolute value of the ζ-potential to an effective "discharge" of the AuNPs.

The phenomenon has been observed for several batches of AuNPs but the time-scale of the aggregation is quite long and the kinetics seem to depend on several variables such as the starting ζ-potential of the AuNPs, and the size of the bulk gold. A plausible explanation is that meanwhile the bulk object captures the electrons previously on the NPs, the driving forces of the discharge decreases because the electrostatic energy of the electrons on the

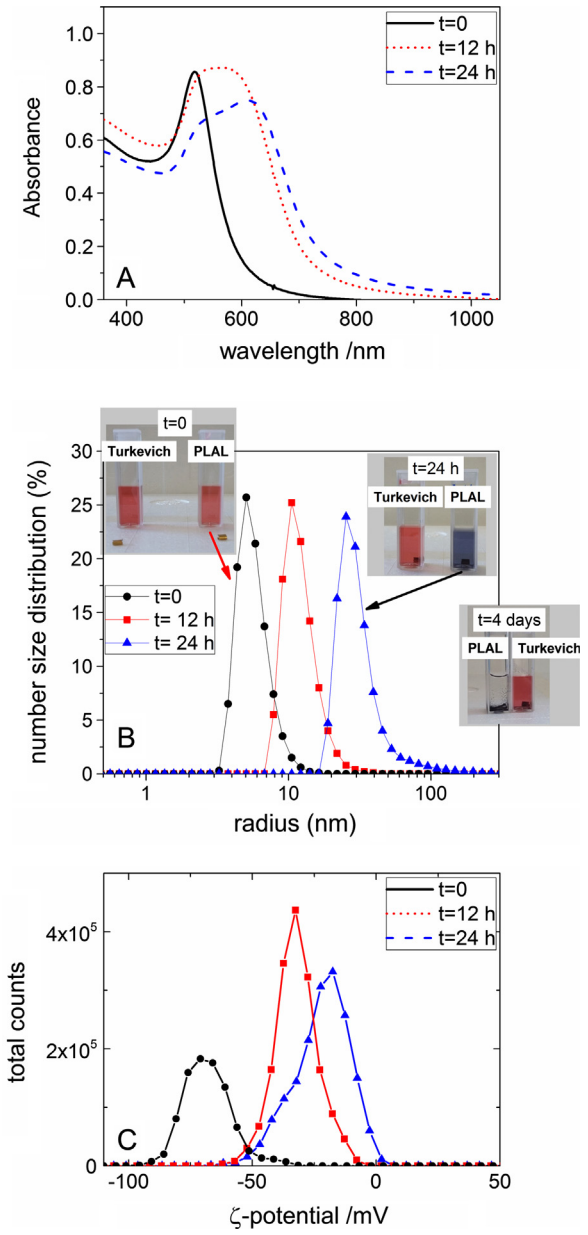


Fig. 6. Effect of the addition of a macroscopic gold object on the solution of AuNPs prepared by PLAL in KCl 10^{-4} M. (Panel A) Absorbance spectra. (Panel B) Number size distribution. (Panel C) ζ-potential distribution. In all the panels the data before the addition are in black, after 12 h in red and after 24 h in blue. Also shown in panel B are pictures before the addition (t = 0, note the gold pieces standing in front of the cuvettes), after 1 and 4 days in the case of solutions of AuNPs prepared by PLAL and by chemical route. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

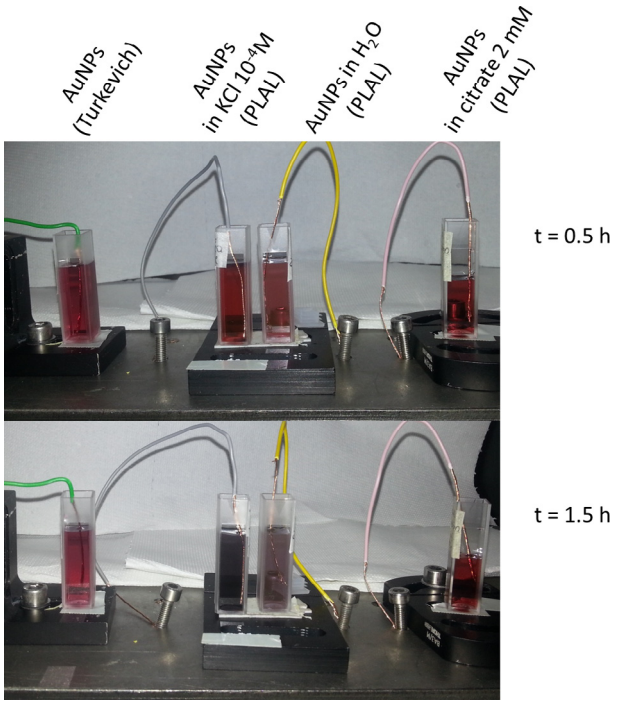


Fig. 7. Setup of the ground wiring experiment, from the left there are AuNP s synthesized via: Turkevitch chemical process, PLAL in KCl 10^{-4} M, PLAL in milli-Q deionized water and PLAL in citrate 2 mM. The upper picture refers to 0.5 h after the grounding; the lower picture refers to 1.5 h after the grounding.

bulk increases (yet remaining much lower than when the electrons were on the AuNPs). If this is the case, ground-wiring the macroscopic metal should speed-up the aggregation process. In addition this should furnish a fair protocol to compare the behavior of different AuNPs. Following such a line of thought, we have dipped a copper wire in the solutions of AuNPs synthesized under different conditions; viz. by means of wet chemical synthesis according to the Turkevitch protocols, and by PLAL of a gold target immersed in: deionized water, KCl 100 μM, and in a solution of citrate 2 mM. The latter procedure of synthesis [17] is quite interesting since it is a classical pulsed laser ablation but performed in an aqueous solution of sodium citrate, a common stabilizer that is known to confer an overall negative charge to the NPs and thus, in the case of NPs synthesized by PLAL is expected to add to the electron-induced stabilization an extra contribute due to the adsorbed citrate molecules.

The protocol for the de-stabilization is described in Fig. 7; briefly the AuNPs synthesized in different media have been ground-wired using copper wires to the same "ground" (strictly the huge metallic mass of the optical table we use for the arrangement of laser and optics in the ablation procedure). This has been done at the same time for the four different solutions (Au NPs prepared by wet chemistry and PLAL in deionized water, KCL, sodium citrate). Two representative pictures of the overall arrangement are shown in Fig. 7.

The first photograph describes the situation few minutes after dipping the "grounded" copper wire while the second gives an unambiguous description of what happens after more than one hour from the introduction of the conductive wire into the AuNPs. It is clearly visible that, once "grounded", the solutions of AuNPs synthesized by PLAL in pure water and in dilute solution of KCl turn dark and start to precipitate in less than two hours while the solutions of AuNPs stabilized by the presence of citrate ions do not.

The time evolution of the system has been quantitatively monitored by withdrawing portions of the solution and measuring the corresponding visible spectrum and the ζ-potential. Fig. 8 shows the behavior of the AuNPs prepared by ablation in absence of citrate (for the sake of clarity only the data for AuNP ablated in deionized water are shown).

It is clear from Fig. 8A that, after a short lag time, the shape of the SPR changes and a new plasmon band grows in the red region of the spectrum at the expense of the original SPR at 520 nm. The change in the shape of the spectrum is almost completed in two hours and for longer times the AuNPs precipitate with a corresponding decrease of the overall absorbance. The kinetics of the process is well described by the evolution of the ratio of the absorbance values collected at 520 nm and 650 nm and can be compared with the evolution of the ζ-potential values (see Fig. 8B).

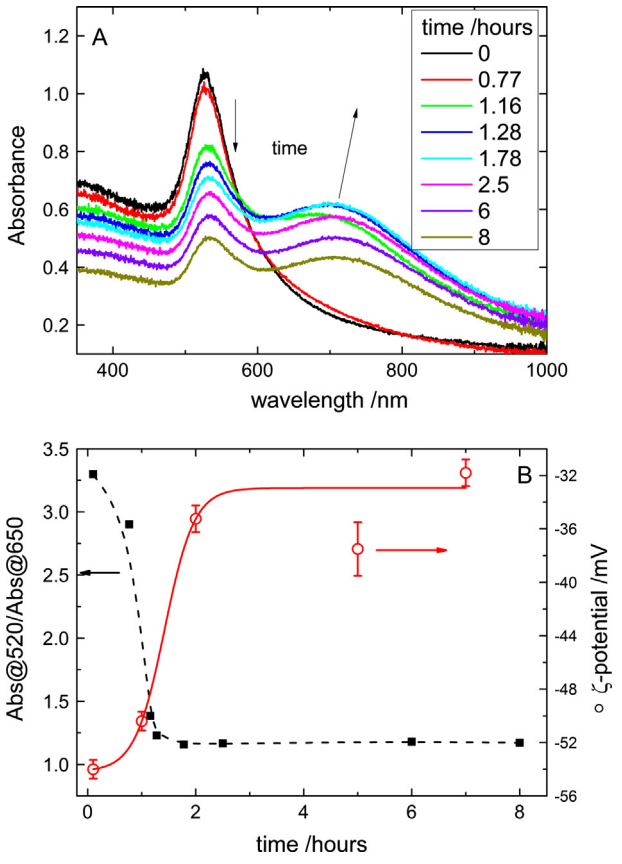


Fig. 8. Effect of ground-wiring on the AuNPs synthesized by PLAL in deionized water. (Panel A) Absorbance spectra at different times from the grounding. (Panel B) Ratio of the absorbances at 520 and 650 (black points left ordinate) and ζ-potential (red open circles, right ordinate) as function of the grounding time. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

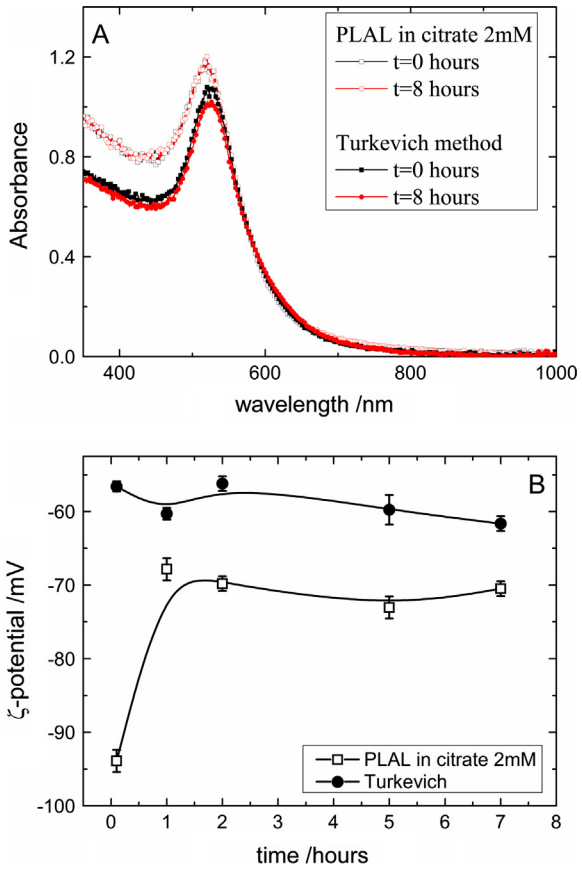


Fig. 9. Effect of ground-wiring on the AuNPs synthesized in presence of citrate; in both the panels closed dots refer to AuNPs synthesized via the Turkevitch chemical process while open dots refer to AuNPs synthesized by PLAL in citrate 2 mM. (Panel A) Absorbance spectra at time = 0 and after 8 h. (Panel B) ζ-potential as function of the grounding time.

In such a case the formation of larger aggregates is parallel to an evolution of the ζ-potential that becomes less negative by 22 mV passing from the original −54 mV to about −32 mV upon "grounding". The latter value is not enough to assure a repulsive interaction sufficient to preclude the aggregation because the large Hamaker's constant of gold. A confirmation that ground-wiring the solutions of AuNPs prepared by PLAL removes the extra electrons from the metallic particles is given in Fig. 9 where is shown the behavior of the AuNPs ablated in citrate solution. The AuNP ablated in presence of citrate have a very negative ζ-potential (−92 mV) likely because in addition to the charge of the trapped electrons they have also absorbed negatively charge citrate ions. Introduction of a grounded metallic wire also in such a case causes the ζ-potential increases by 22 mV reaching −70 mV. Interestingly, the ζ-potential of the citrate-stabilized AuNP synthesized by chemical route is totally unaffected by the presence of the grounding wire (see Fig. 9B).

As a whole, the set of experiments described above are consistent with the hypothesis of an electron-induced negative charge on the AuNPs prepared by PLAL and are hardly compatible with a stabilization due totally to the presence of deprotonated gold oxides or adsorbed anions.

5. Conclusions

In this work AuNP have been prepared by means of pulsed laser ablation in solutions at different pH. Under all the conditions the size of the particles was close to 5 nm in radius. The main effect of the pH of the solution was on the measured ζ-potential that becomes more negative by roughly 30 mV passing form pH = 3 to pH = 10. This in contrast with what one expects for NPs stabilized by gold oxides that, having pKas close to 4, should have a positive ζ-potential at pH < 4. Furthermore, AuNP ablated in solutions of acids with very different oxidation power, viz. HCl, H_2SO_4, HNO_3 (all at pH = 3) have size and ζ-potential that are indistinguishable within the experimental uncertainty. This last observation suggest that the oxidation of the gold surface has only a minor role (if any) on building the negative surface potential. Since the anions Cl^-, SO_4^{2-}, NO_3^- have very different propensity to adsorb to the surfaces (they occupy different positions in the Hofmeister's series) our experimental observation indicates that also the anion adsorption to the gold surface has a negligible impact on the negative surface potential. Having ruled-out surface deprotonation and anion binding as the origin of negative charges on the AuNPs we have focused our attention on the possibility that excess electrons formed within the plasma phase could charge the metallic particles. In this respect, the colloidal size of the NPs is very important:

(i) the particles are so small that small number of excess electrons (5–10) is enough to build a strongly negative ζ-potential that assures the stability against aggregation.
(ii) on the other hand the particles are large enough to reduce the Born energy of charging to a level that doesn't allow direct reaction with the hydrated H^+ present in solution.

Such a heterodox mechanism of stabilization of the metallic particles prepared by PLAL has been tested by adding to the system elements that can remove the putative excess electrons. In particular we have studied the evolution of the AuNP upon the addition of macroscopic bulk objects (that, in principle, should drain the electrons from the AuNPs) and upon contact with a "grounded" copper wire. In both cases the AuNPs lose their charge and aggregates a response that is quite obvious in the case they are stabilized by electrons and very difficult to explain if the stabilization would due to surface oxidized groups or absorbed anions.

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